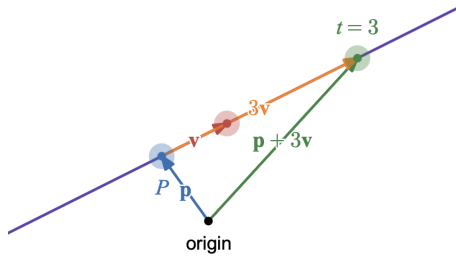


A. **Lines.** Let us use vectors to describe straight lines.



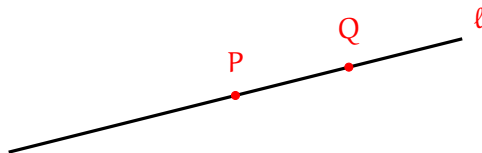
Let  $\ell$  be a line passing through point  $P$  with nonzero direction  $\vec{v}$ .

A **parametrization** for  $\ell$  is given by the function:

$$\vec{r}(t) =$$

where we call  $t$  the **parameter**.

**Example 1.** Consider the line  $\ell$  through the points  $P(0, 4, 2)$  and  $Q(-3, 2, -7)$ .



(a) Parametrize  $\ell$ .

**Solution.** We select a point on the line:

and a nonzero direction for the line:

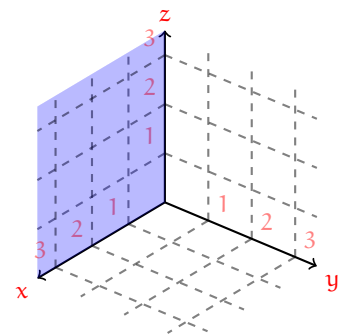
whence the parametrization is:

(b) Find the point where this line intersects the  $xz$ -plane.

**Solution.** The  $xz$ -plane is defined by the equation:

Thus, we set the  $y$ -component of our parametrization to:

And then, substitute this value of the parameter into the parametrization.



**Example 2.** Are the lines with the following parametrizations parallel?

$$\ell_1 : \vec{r}_1(t) = \langle 8 - 8t, 4 - 3t, 2 + t \rangle$$

$$\ell_2 : \vec{r}_2(s) = \langle 4s, 1 + 1.5s, 8 - 0.5s \rangle$$

**Solution.** We observe that the direction vectors can be read off as the:

Thus, the direction vectors for the lines are:

$$\vec{v}_1 =$$

$$\vec{v}_2 =$$

The lines will be parallel if and only if the direction vectors are parallel.

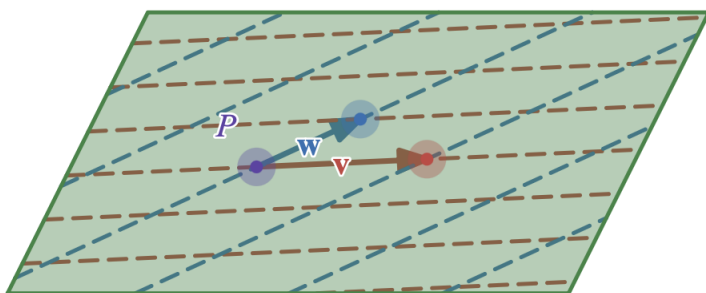
For two direction vectors to be parallel, one vector must:

If  $\lambda$  (“lambda”) denotes a scalar, by investigating the equation:

These equations are simultaneously solved by:  $\lambda =$

Thus the directions, and hence the lines, are parallel.

**B. Planes.** The 2D version of a line is a plane.



•  
origin

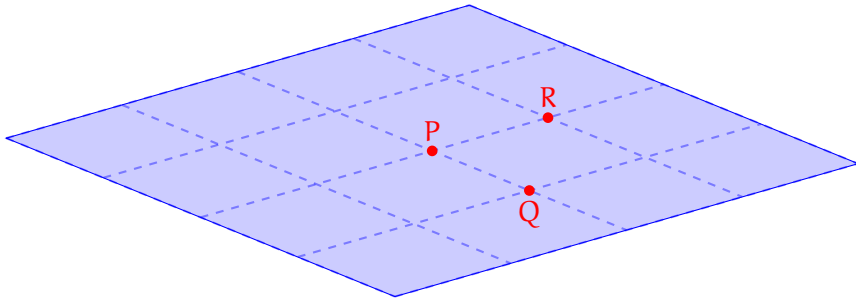
Let  $\mathcal{P}$  be a plane passing thru point  $P$  with nonparallel directions  $\vec{v}$  and  $\vec{w}$ .

A **parametrization** for  $\mathcal{P}$  is given by the function:

$$\vec{X}(s, t) =$$

where we call  $s$  and  $t$  the **parameters**.

**Example 3.** Parametrize the plane that passes through the points  $P(1, -2, 1)$ ,  $Q(-4, 5, -8)$ , and  $R(7, -8, 7)$ .



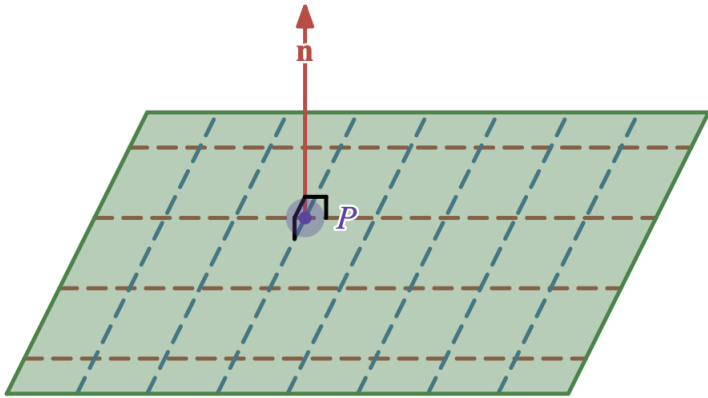
As before this sketch was made without a thought to correctness of orientation and perspective. I nonetheless hope against hope that it will still be of use. Spoiler: it is. The main way it could have failed us is if the three of  $P$ ,  $Q$ , and  $R$  were on a singular line. But of course, then they would not determine a plane.

**Solution.** We select a point on the line:

and two nonparallel directions for the plane:

whence the parametrization is:

C. **Normal Vectors.** There is another method for describing planes in 3D space.



Let  $\mathcal{P}$  be a plane passing thru point  $P$  with nonzero **normal** vector  $\vec{n}$ .

The condition for  $\vec{x} = \langle x, y, z \rangle$  to be the position of a point on the plane is:

$$\vec{n} \cdot \vec{x} =$$

We call this equation a **scalar equation** for the plane.

Let  $\mathcal{P}$  have normal  $\vec{n} = \langle 1, 2, 3 \rangle$  and pass through point  $P(1, 1, 1)$ .

Then its scalar equation is:

The scalar equation of a plane can always be put in form:

in which case  $\langle a, b, c \rangle$  is a normal vector to the plane.